

DEVELOPMENT OF AUTOMOTIVE AIRCONDITIONING TRAINER

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Master of Arts in Technology and Livelihood Education

Major in Industrial Arts

by

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CERTIFICATION

This thesis entitled “**Development of Automotive Airconditioning Trainer**” prepared and submitted by **Brandon A. Libed**, in partial fulfillment of the requirements for the degree **Master of Arts in Technology and Livelihood Education, major in Industrial Arts**, has been examined and is recommended for the acceptance and approval for Oral Examination.

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B.A.L.

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Abstract

This study aimed to develop an Automotive Air-conditioning Trainer intended to enhance instruction and hands-on learning in automotive air-conditioning servicing for students enrolled in AUT 127 at the University of Northern Philippines. Utilizing a descriptive development research method, the study assessed the trainer's functionality, durability, dependability, and safety. Ten (10) expert evaluators provided feedback using a validated questionnaire. The trainer incorporated essential automotive air-conditioning systems, including a mechanical and electrical system that supports real-world application through interactive simulations. Evaluation results showed "Very Much Acceptable" ratings across all parameters, with particularly high marks for dependability and functionality. The trainer was found to have high-quality materials and is well-suited for both classroom and laboratory use at the College of Technology. It is therefore recommended that the enhanced automotive air-conditioning trainer have additional features on fault indicators on the electrical system, cooling system, and control user interface fault. Additionally, the trainer may be utilized by instructors and professors as a standard instructional tool for teaching automotive air-conditioning systems. Future studies may be conducted to evaluate its effectiveness and acceptability for students and teachers.

Chapter I

THE PROBLEM**Introduction**

The automotive industry is undergoing rapid development, with vehicles transitioning from purely mechanical to high-tech systems that incorporate electronics and automated technologies. Due to this development, the qualifications of automotive mechanics are increasingly complicated and demanding. The old way of teaching is not enough to prepare students for the real world, so other, more interactive methods of learning are also required.

Many educational institutions have responded with technology-enabled learning environments that promote engagement and deeper knowledge. Nowadays, industry-based learning environments are being embraced by a variety of educational institutions, enabling active participation and helping learners gain a deeper understanding of concepts. There is also evidence to support this shift, which suggests that there is a direct relationship between the application of technology to teaching and learner engagement, efficiency, as well as the retention of knowledge, with skill-based education as a special example of where theory has to be translated into applied knowledge in the field (Akintayo et al., 2024)

It is a major advance that underscores the need for automotive students to develop relevant competencies, streamline instructors' instructional processes, and encourage the establishment of innovative instructional policies aligned with industry demands.

The Automotive Air Conditioning (AAC) system is one of the most important and complex vehicle systems, with many applications. AAC has evolved from being solely about passenger comfort; it now extends beyond the passenger compartment and is essential for ensuring safe cabin conditions, clean air, and more efficient systems. As a result, electronic control units, variable-displacement compressors, and eco-friendly refrigerants are introduced, all of which require expertise. Despite this, numerous technicians and students are having difficulties understanding the conceptual knowledge of these principles and, accordingly, tend to act incorrectly. According to industry standards, incorrect oil charging and misdiagnosis often result from failing to account for the theoretical basis, leading to system failure (Wang et al., 2019). Both students' conceptual understanding and practical skills must be improved so they can identify a diagnosis and properly service the system, thereby enhancing system reliability in realistic automotive environments. In the Philippines, technical education institutions are subject to technical competency requirements published by the Technical Education and Skills Development Authority (TESDA) for programs such as Transport Refrigeration and Air Conditioning Servicing NC II. However, despite these standards, limited instructional resources and a lack of laboratory equipment remain challenges for most institutions. Such limitations reduce students' opportunities to participate in service activities, leading to shallow learning and low confidence in performing the service. Local studies have shown that the lack of adequate training tools hinders students' acquisition of manipulative and technical skills (Bajet Jr. et al., 2014; Guzman et al., 2015). Addressing this situation is essential not just for technical education but also to ensure that students

not only achieve certification but also possess the proficiency, confidence, and skills necessary to do so in the working world.

In practice, instructional trainers may help close the gap between theory and practice in professional technology education. Contrary to traditional modes of instruction that rely on lectures and static visual aids, trainers offer a more interactive, hands-on learning experience that promotes better comprehension of dynamic systems.

Local trainers have been found to lower learners' cognitive load and connect theory with the practical aspects of systems (Baldos & Chavez, 2025). Additionally, traditional teaching methods are often inadequate for describing complex automotive systems (Hernández-Chávez et al., 2021). These findings are of great importance, as they emphasize the need for new learning tools to encourage meaningful learning, greater student involvement, and academic improvement across the field of automotive education.

With realistic but limited materials, it is impossible to use physical vehicles for teaching purposes. There is also difficulty in physically observing students when space is insufficient, when hidden elements are present, or when a direct view of the whole system is unavailable. This makes learning passive, rather than interactive. To back this up, a study noted significant barriers to vehicle-based instruction in group demonstrations and student participation (Wang, 2025).

In contrast, trainer-based systems afford learners a controlled, accessible setting to observe, manipulate, and analyze elements more effectively (Fitrianto & Saif, 2024; Carisosa et al., 2025). This is important because it enhances students'